

Dylan Bruce

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Education

Georgia Institute of Technology

B.S. in Computer Science, Minor in Mathematics

Expected May 2027

Atlanta, GA

◦ **GPA:** 3.92/4.00

◦ **Activities:** Big Data Big Impact @ GT, WebDev @ GT, Boulderling, Table Tennis

Experience

Software Engineer II Intern

Walmart Global Tech

May 2026 – Aug. 2026

Bentonville, AR

- Reduced average system-monitoring time by **50%** by deploying **2 containerized microservices** to centralize operational data in a unified dashboard
- Consolidated **3 alert sources** into a unified system-health dashboard using Next.js and Spring Boot, giving engineering leadership across **4 subteams** a single view of operational issues
- Automated monitoring of **30+ repositories** and **5 data pipeline DAGs** through configurable collectors with **5-minute polling** and a 3-tier Fresh/Aging/Stale signal classification system

Software Engineer Intern

MathWorks

Jan. 2026 – May 2026

Natick, MA

- Expanded C++ test coverage by **1%** across a **multi-million-line** legacy codebase, using CMake-based CI/CD workflows to investigate and resolve test failures
- Created and published **12+ GenAI "skills"** to guide test generation, improve test buildability, and align with internal testing conventions
- Achieved a **75% test-generation success rate** by designing agentic GenAI workflows with iterative generate-build-fix loops to automatically diagnose and correct test failures

Machine Learning Researcher

Georgia Tech VIP Aquabots: Marine Robotics

Aug. 2025 – Present

Atlanta, GA

- Improved InceptionV4 CNN accuracy from **50% to 82%** for plankton classification across **4,000+ images and 50 species** by enhancing preprocessing and normalization pipelines for underwater imagery
- Collaborated on a **5-person multidisciplinary team** to design and evaluate computer vision preprocessing pipelines, systematically testing image transformations to improve downstream CNN performance

Teaching Assistant - Linear Algebra / Discrete Mathematics

Georgia Tech Department of Mathematics

May 2025 – Present

Atlanta, GA

- Led instruction and review for **1,000+ students** in Linear Algebra and Discrete Mathematics, covering proofs, vector spaces, matrices, and algorithmic complexity
- Mentored students through rigorous problem-solving and mathematical reasoning in weekly instructional sessions

Projects

Pokémon Go Information Search Platform (pogo-db.com)

Jul. 2026 - Aug. 2026

- Built a Next.js data-driven information platform serving precomputed combat data across **1,000+ entities** using a Python build-time data engine
- Standardized unstructured game mechanics by normalizing raw data across **3+ disjointed web resources** using an automated scraping and ETL ingestion pipeline into PostgreSQL.
- Automated **100%** of live event updates by building scheduled CI/CD workflows that re-trigger build-time ETL pipelines and dynamic deployments.

FishCast - Climate-Fishery Forecasting

Sep. 2025

- Developed **LSTM-based** forecasting models in **PyTorch** to predict temporal shifts in marine species distributions from integrated climate and fisheries datasets
- Trained ML models using Georgia Tech PACE **supercomputing resources**, collaborating on a **5-person multidisciplinary team** to develop and benchmark forecasting approaches

Technical Skills

Languages: Python, C++, C, Java, C#, JavaScript, TypeScript, SQL, MATLAB

Frameworks: React, Next.js, Spring Boot, TensorFlow, PyTorch, scikit-learn, Redux

Infrastructure: Docker, Kubernetes, AWS, Jenkins, Apache Airflow, Azure Cosmos DB, Prometheus

Tools & Databases: Git, GitHub Actions, CMake, Maven, PostgreSQL, Firebase, SonarQube

ML/Data: NumPy, Pandas, Matplotlib, XGBoost